

		BAHIR DAR UNIVERSITY BAHIR DAR INSTITUTION OF TECHNOLOGY (BiT) FACULTY OF MECHANICAL AND INDUSTRIAL ENGINEERING							
Course Guidebook									
Course Title:		Manufacturing Engineering I			Chair Name:		Manufacturing Eng.		
Course Code:		MEng. 3192							
Module:		Manufacturing Engineering							
CPs:		5							
Contact Hours:		Lecture		Tutorial		Laboratory or practice		Home study	
		2		3		2		3	
Target Group		3 rd MED							
Academic Year		2024/2025							
Semester		2 nd							
Instructor's Name		Gessesew L.							
Instructor's Address		Office No.		E-mail		Consultation hour			
		NDD8-2;		gessesew2002@gmail.com		Every Friday After 4:00PM			

Learning Outcomes:

The course enables the students to understand:

- Basic concept of casting process, design of cast, casting defect and their remedies;
- Basic principles and mechanisms of shearing and metal-forming process of selected processes;
- Material consumption, forces and work done on selected machines and die design;
- Basic principles and mechanisms of (plastics, rubbers, polymer matrix composite, ceramic and glass forming) processing;
- Basic principles of powder metallurgy,

Course Description:

Basic concepts of manufacturing process, casting process, casting design, casting defect and their remedies.
Fundamental of shearing and metal forming process.
Principles of plastic, rubber, composite, ceramics and glass manufacturing process.
Fundamentals of powder metallurgy.

Course Content:

Week	Topics	Methodologies	Lab. Activities	References
1	1. Introduction about different manufacturing processes.	Lecture		Text: Page: 1-19
2-3	2. Introduction basic casting processes (Expendable and Permanent molding)	Lecture, Tutorial, Quiz 1	Demonstration of casting equipment;	Text: Page: 205-224; 225-230
4	3. Design of different cast elements: Pattern, Mold, Riser and Gating system	Lecture, Tutorial, Assign 1	Pattern design, pattern making, Aluminum melting, and pouring.	Text: Page: 210-220
5	4. Casting defects and their remedies,	Lecture	Examine the defect that occurred on the cast product during lab exercises.	Text: Page: 249-251
5-7	5. Bulk deformation processes): (Rolling; Forging; Extrusion; Drawing; miscellaneous forming processes)	Lecture Tutorial		Text: Page: 383-442
8	6. Sheet metal forming and shearing: (Cutting, Bending and Drawing process); Sheet metal dies,	Lecture Tutorial Quiz 2	Demonstration on sheet metal equipment's and making simple product from sheet metal	Text: Page: 443-461
9	7. Fabrications of plastics, Ceramics and Composite	Lecture		Text: Page: 268-314; 327-345;
10	8. Powder Metallurgy: (Powder manufacturing, Mixing, Blending, Compacting and Sintering), Design considerations in powder metallurgy	Lecture Assign. 2		Text: Page: 344-367
	Final Exam			



Teaching and Learning Methods

This course will incorporate a combination of **lectures, tutorial and laboratory demonstration**. Active participation of the students in all class activities are very essential and encouraged to accomplish the goals of the course.

Assessment Methods	
Assessment Type	Assessment Weight (%)
Individual Assignments (I&II)	15 (7.5+7.5)
Quizzes (I&II)	10 (5+5)
Mid exam (Intermediate Exam.)	25
Final Exam	50
Total Mark	100%

Course Policy

Minimum of 85% attendance during lecture hours, and 100% attendance during practical work and tutorial sessions, except for some unprecedented mishaps. All students are expected to abide by the code of conduct of students on BDU, BiT Legislation.

Text Books:

Mikell P. Groover; Fundamentals of Modern Manufacturing; John Wiley & Sons, Inc.; 2004;

References:

1. Serop Kalpak Jain & Steven R. Schmid, Manufacturing Engineering and Technology (4th Edition),
2. Hwaiyu Geng, Manufacturing Engineering Hand Book, Mar 1, 2004;
3. James G. Bralla, Handbook of Manufacturing process-How Product Components and materials are made, Jan 15, 2007;
4. John A Schey, Introduction to Manufacturing processes (McGraw-Hill Series in Mechanical Engineering & Material Science), Mar 1, 2000;
5. Winkelmann, Manufacturing Engineering (Teaching materials), Technical University of Dresden, 1982
6. Beddoes J., Principles of Metal Manufacturing Processes, John Wiles & Sons Inc, New York, 1999
7. Rao P.N., Manufacturing Technology, Second edition, Tata McGraw-Hill Publishing Company Limited, New Delhi, 1998

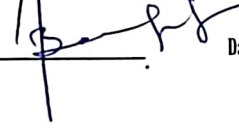
Authorization:

Prepared instructor's Name: Gessesew Likeleh

Sign. 

Date: Feb. 18, 2025

Verified chair holder's Name: Assela Asemare(Ph.D.)

Sign. 

Date: Feb 18, 2025

